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AI-driven environmental sensor networks and digital platforms for urban air pollution monitoring and modelling

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ABSTRACT

Recent advances in Artificial Intelligence (AI) research have opened up new opportunities for leveraging AI research for societal impacts. AI research offers novel ways of tackling societal problems including environmental, health, and education challenges. Despite the potential, there are limited documented use cases and methodologies for translating AI research to societal impact at a large scale. This paper presents AirQo, an AI and advanced technology-driven use case for urban environmental pollution monitoring and modelling and the resulting societal impacts that have been realised. The research outputs include a set of digital solutions for the environmental air pollution challenges including (1) custom-designed low-cost air quality monitors that are premised on IoT technology (2) a methodology for deploying a high-resolution and citizen-driven air quality monitoring (3) AI-powered digital tools for air quality information modelling and analysis for citizens and city leaders, and (4) a framework for engagement for citizens and leaders. The AirQo project has been deployed and scaled out in cities in Eastern, Western, and Central African countries. The societal impacts resulting from the implementation of the AirQo research project include policy and regulations, education and awareness, and research around air quality issues.

SPECIFICATIONS TABLE

Subject area	Please select **one** CATEGORY for your manuscript from the list available at: All Science Journal Classification Codes
More specific subject area	Briefly describe the narrower subject area. Max 150 characters
Category/categories of societal impact	Please select category or categories from **dropdown list** (you can select multiple categories): Cultural Economic Education Environmental Health Legal Political Societal Technological

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Sustainable Development Goals (SDGS) the research contributes to	Please select SDGs from **dropdown list**. You can select multiple SDGs. *GOAL 1: No Poverty* *GOAL 2: Zero Hunger* *GOAL 3: Good Health and Well-being* *GOAL 4: Quality Education* *GOAL 5: Gender Equality* *GOAL 6: Clean Water and Sanitation* *GOAL 7: Affordable and Clean Energy* *GOAL 8: Decent Work and Economic Growth* *GOAL 9: Industry, Innovation and Infrastructure* *GOAL 10: Reduced Inequality* *GOAL 11: Sustainable Cities and Communities* *GOAL 12: Responsible Consumption and Production* *GOAL 13: Climate Action* *GOAL 14: Life Below Water* *GOAL 15: Life on Land*

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	GOAL 16: Peace and Justice Strong Institutions *GOAL 17: Partnerships to achieve the Goal*
Resource availability	*If applicable, include links to resources showcasing impact of research (e.g., data, software, hardware).*
Related research article	*If your article is related to a research article - please cite your associated research article here.* ***For example:*** *J. van der Geer, J.A.J. Hanraads, R.A. Lupton, The art of writing a scientific article, J. Sci. Commun. 163 (2010) 51–59. https://doi.org/10.1016/j.Sc.2010.00372*
Stage of research	*Please indicate whether this paper outlines the predicted/intended societal impacts of a project or demonstrates the societal impacts of a project.*

Social impact

Environmental air pollution is a major public health challenge that affects urban spaces across the world. Previous studies have reported that more than eighty percent of the population living in urban spaces where air quality is monitored breathe polluted air [1]. There is increasing scientific evidence on the negative impacts of air pollution on human health, economics, and the environment. The World Health Organization (WHO) estimates that 6.7 million people die every year from illnesses related to exposure to air pollution, with countries in the low- and middle-income among the most impacted [2]. Rapid urbanisation together with population growth are expected to exacerbate the air pollution challenges in cities [3]. The WHO provides guidelines on the expected air quality standards [4]. In low- and middle-income countries, ninety-eight percent of cities are estimated not to meet the WHO air quality guidelines [1]. Addressing the air pollution challenge directly contributes to the progress on the attainment of the United Nations Sustainable Development Goals (SDGs) specifically: "By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination" (SDG-3.9), and, "By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management" (SDG 11.6) [5].

Internet of Things (IoT) and artificial intelligence (AI) research and systems provide promising means to accurately track and understand spatial and temporal variations of air pollution, thereby empowering decision-makers and citizens with timely information to undertake evidence-informed actions to reduce or better manage the air pollution challenge. Designing and setting up a distributed large-scale IoT and AI system for environmental challenges in low-resource settings is non-trivial. Many technical and practical challenges exist, to wit, data transmission, network powering, data management, physical and system security, running and maintenance costs, and building and deployment of AI models for use in resource-constrained settings [6]. Examples of research outputs and data systems for such use cases are few or non-existent in many low- and middle-income cities, making it difficult to understand the trends and scale of air pollution [7], contributing to the inaction and lack of progress in many cities. The cost of setting up sustainable air quality data systems is a major challenge for many cities leading to the dearth of evidence and data on the scale and magnitude of air pollution in cities [8,9,6]. In the previous work, we describe the cost and other design considerations [6,10].

In this paper, we present IoT and AI-driven digital solutions and the societal impact of the AirQo[2] research project. The AirQo research project was conceived to support African cities to tackle environmental air pollution using custom-built and locally developed digital technologies and computational intelligence tools. The AirQo research project includes a set of digital solutions [6] for the environmental air pollution challenges including (i) custom-designed low-cost air quality monitors that are premised on IoT technology (ii) a methodology for deploying a high-resolution and citizen-driven air quality monitoring (iii) AI-powered digital tools for air quality information visualisation and interpretation for citizens and city leaders, and (iv) a framework for engagement for citizens and leaders. The AirQo project has been deployed and scaled out in cities in Eastern, Western, and Central African countries. We present the societal impacts resulting from the implementation of the AirQo research project in the areas of policy, community education, education and awareness.

AirQo is a societal impact research project started in 2015 at Makerere University, Kampala, Uganda with a focus on using IoT, data and AI technologies to support cities in Africa to establish sustainable and low-cost data systems for tracking and managing air quality whilst empowering communities and authorities with timely air quality data to inform action. The project has evolved through different phases of research and technology conceptualization, design, development and deployment, research translation, and impact. The project has been scaled to over ten cities in eight African countries in Eastern, Central and Western Africa (Fig. 1). Table 1 provides the family of the data and AI products in the AirQo research project. For each product, we outline what it is, who benefits, and how they benefit.

Methodology

Digital solution development methodology

AirQo's methodology for developing robust digital solutions is grounded in a user-centred design (UCD) approach [11,12] and agile development methodology [13]. This ensures that both the software and hardware systems are designed to meet user needs and adapt to operating and deployment contexts in the target settings of cities in low- and middle-income countries. The development of technology products is derived from and driven by a deep understanding of user requirements and deployment environments. For instance, air quality monitors are designed to be installed on both static objects like electric poles, and buildings, and mobile objects like motor-cycle taxis. Data transmission is optimised for cellular connectivity, which is prevalent in most African urban areas, while WiFi is given lower priority [6]. In this methodology, collaboration plays a vital role throughout this development process. The digital air quality analytics platform and dashboards are co-created with city authorities, ensuring their relevance and effectiveness. The integration modelling and analysis results in the air quality mobile app are done in close collaboration with the intended end-users, guaranteeing a user-friendly experience. The research project employs the agile development methodology, particularly the scrum project management system [14]. By using tools like Jira, Slack, and Figma, the team integrates the Objectives-Key-Results (OKR) system [15] into sprints to achieve short-term project goals with the end-users' needs in mind. OKRs project goal and results setting and progress tracking framework is used to improve productivity, knowledge sharing, and transparency among teams, particularly in the era of distributed teams and remote work [16].

Stakeholder and community engagement methodology

The AirQo stakeholder and community engagements are premised on inclusive and co-design methodology [17]. For a given city, the project starts with stakeholder mapping and analysis as a preliminary means to understand the stakeholder needs, roles, levels of interests, and influence. Initial engagements focus on cultivating strategic partnerships with mandated city authorities, government ministries, local

[2] https://airqo.africa

Fig. 1. Spatial map of the AirQo air quality digital platform in Africa.

governments, and relevant agencies. Early efforts involve alignment with the local governance structures, to generate a sense of commitment and ownership, ultimately contributing to more effective and lasting research uptake for air quality management practices in the cities. This approach is geared towards driving research impact through policy action and ensuring the successful adoption of the research output in a given city or geographical area of focus. The project uses focus group discussions to incorporate the voices of grassroots communities and key stakeholders in the study design and project during the initial stages. To ensure inclusivity, the focus group discussion involves local individuals, environment, transport, and health representatives as well as key authorities from the local councils like the environment officers. The discussions are primarily in their local languages and held in community centres like schools, community halls, and playgrounds. Some members of the community are identified as air quality champions who champion the uptake of the research products and knowledge among their communities. Community champions are selected from the focus group discussions based on the knowledge and interest in utilising data products to educate other community members. We use the focus group discussions to identify potential community champions representing sectors. In some cases, community champions may already hold roles within their communities, for example, environmental officer, local leader, or member of Village Health Team. The champions take on the role of educating and raising awareness about air quality and air pollution in their communities.

Research impact assessment methodology

Assessing research's societal impact requires a different methodology from the traditional scholarly impact metrics such as the number of publications, citations, impact factor, and the h-index [18]. We used the Outcome Harvesting approach [19,20] to map out and assess the potential impacts and outcomes from the AirQo project. The outcome harvesting methodology provides a systematic framework for documenting and validating outcomes from a social impact project. The outcomes include changes and impacts that have happened to institutions and significant changes to individuals as the result of the research project. Changes at institutions include adoption and use of the results in decision making, policy, and resource prioritisation. Changes at individual level include changes in behaviour and adoption of new practices. We organised workshops and focus group discussions including selected representatives of institutions and individuals that have been impacted on by research results from the project. Institutions include cities and municipalities, environmental regulatory agencies, government ministries, universities, cultural institutions and civil society. Individuals include researchers, students, community champions, local leaders, technical personnel from government departments, and ministries and agencies. These workshops and focus groups enabled data collection on the potential impacts in the form of outcome statements and the verification of the "harvested" outcomes.

Table 1
Types of data and AI research outputs and their pathways to translation, usage, and impact.

Research products	What	Who Benefits	How
Custom low-cost and custom-designed air quality monitors	Low-cost air quality monitor custom-designed for African urban spaces.	Cities, government, and private institutions Individuals/ citizens	Accessible technology for monitoring air pollution Affordable technology Availability of local expertise and capacity
High-resolution environmental sensing network	A high-resolution network of monitors that meets the monitoring goals of a city.	Cities Communities Schools Industries Researchers	Tracking and monitoring of air quality in urban spaces of interest. Near real-time monitoring of air quality at a lower cost.
Network management tooling	Software tools for network management	Network managers Device owners and manufacturers	Enables tracking and management of a large-scale network of air quality monitors.
Calibration methods and tools	Machine learning-based calibration algorithms for low-cost air quality monitors Calibrations software products Documented methods and procedures for quality assurance	Organisations and researchers operating or using air sensor networks Data managers.	Access to high-quality datasets that are quality-assured and calibrated against the reference grade monitors and standards.
Air quality modelling	Spatial modelling of air quality in places without sensors Forecast of air quality	Policymakers Members of the public Schools and other agencies	Authorities can understand large-scale variations of air quality in a given area. Individuals have access to forecast air quality information to plan their activities. Schools and other agencies can use the information to minimise high exposure to air pollution
Air quality analytics platforms	Air quality analytics tools Actionable air quality reports and insights	City leaders Policymakers Journalists	Enabling access to trends and analysis of air quality information. Generating air quality reports for dissemination and decision-making.
Web and mobile products	End-user facing air quality data products Timely and location-specific air quality information	Members of the public Air quality champions Sensitive groups that need to track or minimise air pollution exposures	Access to timely air quality information in locations of interest Actions that can be taken to reduce air pollution Sharing air quality information with other persons
Sensor placement model and tools	AI-powered tool to determine sensor installation places.	City authorities and planners Organisations and individuals setting up and managing air quality networks.	Given a geographical scope, a sensor placement model can suggest the optimal number of required sensors.
Data products and Application Programming Interfaces (APIs)	Air quality datasets available APIs for access to air quality data and other services	Researchers and students Policymakers Developers Agencies e.g., Media houses	Researchers using air quality data for studies e.g., implications of air pollution on health. Developers building third party products Agencies e.g., media houses package the data and broadcast it to a wider audience to raise awareness

Results and implications

This section provides societal impact statements resulting from the AirQo research project for the different categories and target beneficiaries. The research's societal impacts are categorised by Policy, Education and Awareness, and Research.

Policy impacts

The primary target beneficiaries for policy impacts include national environmental regulators, government institutions, government ministries, and city authorities. Air quality datasets, data modelling and insights from AirQo's digital platforms have provided baseline for and contributed the development of regulations and policies at city and national levels. Cities and governments are also able to report and track the state of air quality for inclusion in reports. Previously this was not possible due to a lack of long-term air quality datasets on the air quality across the urban spaces. Technical officers are trained on data access and usage. Further, we packaged the findings showing the trends of air quality in form of policy briefs. The policy briefs include recommendations on the contextualised actions that can be taken to improve air quality. The AirQo researchers were co-opted in the taskforce for drafting the standards. The Uganda National Environment Management Authority started providing the State of Air Quality data, and insights in the National State of Environment report [21]. Further, Air Quality standards at a national level have been developed and are undergoing the stages of approval by the government. At the city level, cities have initiated city-specific air quality action plans, for example, Kampala. African cities like Kampala, Fort Portal, Nairobi, and Kisumu are initiating actions to reduce air pollution including improved waste management, tree planting initiatives, and formulating air quality action plans. At the city level, policy actions can take different forms including a declaration or ordinance by the city council or a local regulation depending on the governance structure in the country. For example, Jinja City in Uganda, formulated clean air city action plans and made a budget allocation air quality improvement.

Education and awareness

The primary target beneficiaries for education and awareness include the public and leaders. Trends analysis of air quality data and state of air quality are made available to the public through AirQo's digital platforms and the application programming interface (API). The AirQo API can be integrated in an entity's website, for example, Kampala, and Nairobi. The digital platforms include the AirQo mobile app,

and the air quality analytics platform. The public is educated and made aware of the dangers of air pollution through our digital social media channels, capacity-building training and campaigns, such as the annual air quality awareness week, joint press conferences with policymakers and year-round community engagements informed by research outputs and findings. This also includes targeted knowledge-sharing with specific agencies, for example, mechanics appreciating air quality management and considering best practices to reduce vehicle emissions in their operations and processes. There are examples of broadcasting air quality updates from the AirQo digital platforms on National Television to increase public education, and awareness of the quality of the air in locations of interest. Further, through our awareness and research outputs communication, influential members of the community and local leaders are advocating for clean air in their communities and neighbourhoods. For example, radio station presenters access the air quality information and use the radio platform to disseminate it to a wider audience. A local leader uses the air quality information to educate communities on the dangers of air pollution to people's health. This has led to communities being aware of the air pollution problems and the actions they can take to reduce exposure or contribute to cleaner air in their communities. Influential members of the community and local leaders advocating for clean air in their communities and neighbourhoods. For example, radio station presenters access the air quality information using the AirQo mobile app and use the radio platform to communicate it to the members of the general population. A local leader uses air quality information to educate citizens about the dangers of air pollution to people's health. This has led to communities being aware of the air pollution problems and the actions they can take to reduce exposure or contribute to cleaner air in their communities.

Academic and research impacts

The primary target beneficiaries include researchers and Academia. AirQo datasets [22] and digital products have enabled studies in the field of air quality modelling and its impacts on health and other issues [23–25]. Students and researchers can access datasets and apply computer science knowledge in the creation of new research products. Another use case of the dataset is machine learning calibration models for a large-scale air quality sensor network [26]. Further, several machine learning and data science challenges based on the dataset have been organised with the aim of capacity building and skilling for students and researchers.

Ethics statements

This work does not involve human or animal subjects.

CRediT authorship contribution statement

Deo Okure: Conceptualization, Data curation, Project administration, Validation, Writing – review & editing. **Maclina Birungi:** Data curation, Methodology, Writing – review & editing. **Angela Nshimye:** Data curation, Methodology, Validation, Writing – review & editing. **Faith Busigu Daka:** Data curation, Methodology, Writing – original draft. **Priscah Adrine Warigo:** Data curation, Methodology, Writing – original draft, Writing – review & editing. **Engineer Bainomugisha:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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